# 2.4 Modeling with Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.4 · Student sheet

**I can…**

- write quadratic equations using vertex form and intercept form.
- use finite differences to tell whether data are linear, quadratic, or neither.
- model real-life data and situations with quadratic functions.

**Vocabulary** · fill in the missing word

**Vertex form** — $y = a(x - h)^2 + k$: use when you _____ the vertex and one more point

**Intercept form** — $y = a(x - p)(x - q)$: use when you know the _____ and one more point

**First differences** — the differences between _____ y-values (equally spaced x)

**Second differences** — the differences of the first differences; constant second differences mean _____ data

## Part 1 · Ball toss data

| Time (s) | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 |
|---|---|---|---|---|---|---|
| Height (ft) | 6 | 22 | 30 | 30 | 22 | 6 |
| First differences | | | | | | |
| Second differences | | | | | | |

Type of function: _____ When is the ball highest? _____ About how high? _____

## Part 2 · Notes and examples (fill in each note, then work the example)

**Which form?:** Know the vertex: y = _____. Know the x-intercepts: y = _____.

### Example 1

Write an equation of the parabola with vertex (2, 5) that passes through (4, −3).

### Example 2

Write an equation of the parabola with x-intercepts −1 and 4 that passes through (2, 12).

**Differences:** Constant first differences → _____; constant second differences → _____.

**Example 3**

Tell whether the data are linear, quadratic, or neither. Write a function for the data.

| x | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| y | 2 | 5 | 12 | 23 | 38 | 57 |

**Example 4**

You have 80 feet of fencing to make a rectangular pen against a barn (the barn is one side). What dimensions give the greatest area? What is that area?

## Part 3 · Partner practice

**Partner 1**

Write an equation of the parabola with vertex (−3, 1) that passes through (−1, 9).

**Partner 2**

Write an equation of the parabola with x-intercepts 0 and 6 that passes through (3, 18).